Optimal Control of a Variable-Mass Thrust-Vectoring Lander*

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***Abstract*— This project studies the optimal guidance and control of a planar variable-mass rocket/lander with translational and rotational dynamics, bounded controls, nonlinear coupling, free-final-time optimal control, endpoint manifold constraints, and trajectory tracking. The project combines Pontryagin's Maximum Principle (PMP), controllability analysis, Linear Quadratic Regulation (LQR), minimum-time optimal control, nonlinear indirect optimal control, and numerical two-point boundary value problem (TPBVP) solvers. The project is inspired by powered-descent guidance and reusable rocket landing systems studied in modern aerospace optimal control literature. All theoretical derivations must be based on PMP. Numerical implementations may use indirect shooting, multiple shooting, continuation or homotopy. methods, and indirect/direct hybrid approaches, provided that the optimality conditions are derived from PMP.**

I. Problem Formulation

The state space of the lander can be represented as:

$$x = \begin{bmatrix} p_x & p_z & v_x & v_z & \theta & \omega & m \end{bmatrix}^\top, \quad u = \begin{bmatrix} \delta T & \tau \end{bmatrix}^\top$$

Symbol	**Description**
State Variables (x)	
p_x, p_z	Horizontal and vertical position
v_x, v_z	Horizontal and vertical velocity
θ	Vehicle pitch angle
ω	Angular velocity (pitch rate)
m	Instantaneous vehicle mass
Control Variables (u)	
T	Commanded thrust magnitude
τ	Commanded control torque

The nonlinear equations of motion governing the lander, $\dot{x} = f(x, u)$, are defined by its kinematics, gravity, thrust resolution, and mass depletion rate.

$$\dot{x} = \begin{bmatrix} v_x \\ v_z \\ \frac{T}{m}\sin(\theta) \\ \frac{T}{m}\cos(\theta) - g \\ \omega \\ \frac{\tau}{I} \\ -\frac{T}{I_{sp} g_0} \end{bmatrix} \tag{1}$$

TABLE I: System Constraints

Mathematical Constraint	**Description**
Control Constraints	
$T \in [T_{\text{min}}, T_{\text{max}}]$	Actuator bounds on thrust magnitude
$\tau \in [\tau_{\text{min}}, \tau_{\text{max}}]$	Actuator bounds on control torque
Numerically Enforced State Constraints	
$p_z(t) \geq 0$	Hard deck limit preventing subsurface flight
$m(t) \geq m_{\text{dry}}$	Minimum mass limit (dry mass)

TABLE II: Suggested System Parameters

Symbol	**Value**	**Description**
g	9.81 m/s^2	Gravitational acceleration
I	10 kg · m^2	Vehicle moment of inertia
m_0	20 kg	Initial vehicle mass
α	0.005 kg/(N · s)	Propellant consumption rate
T_{min}	0 N	Minimum thrust limit
T_{max}	300 N	Maximum thrust limit
τ_{min}	-30 N · m	Minimum torque limit
τ_{max}	30 N · m	Maximum torque limit

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II. Numerical Methodology

A. Linear Systems Solver: The Sweep Method (Parts I & II)

Briefly explain how you used $scipy.integrate.solve_ivp$ (or MATLAB's ode45) to integrate the Riccati equation backward in time to generate the feedback gain matrix $K(t)$. An interpolator was then used on this gain during the forward-pass simulation of the state trajectory.

B. Nonlinear TPBVP Solvers (Parts III & IV)

Acknowledge that the Riccati substitution fails for nonlinear dynamics. Explain the numerical solver you chose for the later phases (e.g., Indirect Shooting using $scipy.integrate.solve_bvp$, or Direct Collocation). Briefly explain how it tackles the coupled state/costate equations.

C. Constraint Enforcement Strategies

- Control Constraints (T, τ): Thrust limits were handled differently for linear and nonlinear phases. For the linear phases, I applied post-calculation saturation (clipping the LQR commanded thrust). For the nonlinear phases, PMP mathematically handles the bounds or how you passed them as boundary arrays to your collocation solver.
- State Constraints ($p_z \geq 0$, $m \geq m_{dry}$): Adjoining state constraints directly to the PMP Hamiltonian creates severe analytical difficulties (jump discontinuities in the costates). The numerical workaround is to either:
  - Apply "Soft Constraints" (adding massive penalties to your Q matrix if the rocket goes underground)
  - "Event Tracking" (terminating the ODE solver upon impact)
  - direct constraint arrays in an NLP solver.

III. Part I — Fixed-Time LQR Around Different Nominal Trajectories

A. Phase Formulation

Linearization: The linearized model plant is formulated as $\dot{x} = Ax + Bu$ where $A = \frac{\partial f}{\partial x}$, $B = \frac{\partial f}{\partial u}$, and $f(x, u) = x$

- Hover

$$A_{\text{hover}} = \begin{bmatrix} 0&0&1&0&0&0&0 \\ 0&0&0&1&0&0&0 \\ 0&0&0&0&g&0&0 \\ 0&0&0&0&0&0&\frac{-g}{m_0} \\ 0&0&0&0&0&1&0 \\ 0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0 \end{bmatrix}, \quad B_{\text{hover}} = \begin{bmatrix} 0&0 \\ 0&0 \\ 0&0 \\ \frac{1}{m_0}&0 \\ 0&0 \\ 0&I^{-1} \\ -\alpha&0 \end{bmatrix}$$

- Descent

$$A(t) = \begin{bmatrix} 0&0&1&0&0&0&0 \\ 0&0&0&1&0&0&0 \\ 0&0&0&0&g&0&0 \\ 0&0&0&0&0&0&\frac{-g}{m^*(t)} \\ 0&0&0&0&0&1&0 \\ 0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0 \end{bmatrix}, \quad B(t) = \begin{bmatrix} 0&0 \\ 0&0 \\ 0&0 \\ \frac{1}{m^*(t)}&0 \\ 0&0 \\ 0&1/I \\ -\alpha&0 \end{bmatrix}$$

$m^*(t) = m_0 e^{-\alpha g t}$

B. Analytical Derivations (PMP)

- Hamiltonian construction

$$H(x, u, p) = L(x, u) + p^\top f(x, u)$$

where $L(x, u) = x^\top Q x + u^\top R u$

$$H(x, u, p) = x^\top Q x + u^\top R u + p^\top (Ax + Bu)$$

- Transversality

$$p(t_f) = \left.\frac{\partial \phi}{\partial x}\right|_{t=t_f}$$

$$p(t_f) = 2Q_f x(t_f)$$

- Adjoint Equations (costate dynamics)

$$\dot{p} = -\frac{\partial H}{\partial x}$$

Using matrix calculus (and the symmetry of Q):

$$\dot{p} = -\left(2Qx + A^\top p\right) = -2Qx - A^\top p$$

- Stationarity Conditions

$$\frac{\partial H}{\partial u} = 0$$

$$2Ru + B^\top p = 0$$

$$u^* = -\frac{1}{2} R^{-1} B^\top p$$

- Riccati Differential equation
Assumption (Riccati Substitution):

$$p(t) = P(t) x(t)$$

Where $P(t)$ is an unknown, time-varying symmetric matrix. Differentiate the assumption. Using the product rule:

$$\dot{p} = \dot{P} x + P \dot{x}$$

Substitute $\dot{p}$ and $\dot{x}$ (the dynamics).

$$-2Qx - A^\top Px = \dot{P}x + P(Ax + Bu^*)$$

Substitute the optimal control u^*. Remembering that $p = Px$, becomes $u^* = -\frac{1}{2}R^{-1}B^\top Px$.

$$-2Qx - A^\top Px = \dot{P}x + P\left(Ax - \frac{1}{2}BR^{-1}B^\top Px\right)$$

$$-2Qx - A^\top Px = \dot{P}x + PAx - \frac{1}{2}PBR^{-1}B^\top Px$$

Group all terms multiplying $x(t)$. Move everything to one side of the equation:

$$\left[\dot{P} + PA + A^\top P - \frac{1}{2}PBR^{-1}B^\top P + 2Q\right]x = 0$$

Because this equation must hold true for any arbitrary state trajectory $x(t)$, the bracketed matrix expression must identically equal zero. Setting the bracket to zero yields the Riccati Differential Equation (RDE):

$$\boxed{\dot{P} = -PA - A^\top P + \frac{1}{2}PBR^{-1}B^\top P - 2Q}$$

- Optimal State-feedback law

$$P(t_f) = 2Q_f$$

$$u^*(t) = -K(t)x(t)$$

$$K(t) = \frac{1}{2}R^{-1}B^\top P(t)$$

C. Numerical Implementation and LQR Tuning

- Q, R and Q_f matrix comparisons. We used heuristic methods to create a selection of Q, R, and Q_f that accurately reflect underlying control intuition.

D. Analysis

Controllability Analysis

- Algebraic Controllability Matrix (Frozen-Time):
  Used to check the structural control authority of the LTI system (or a "frozen" snapshot of an LTV system). For a system with $n = 7$ states:

$$C = \begin{bmatrix} B & AB & A^2B & A^3B & A^4B & A^5B & A^6B \end{bmatrix}$$

- Finite-Horizon Controllability Gramian:

$$W_c(0, t_f) = \int_0^{t_f} \Phi(0,\tau)B(\tau)B(\tau)^\top\Phi(0,\tau)^\top d\tau$$

  Where $\Phi(t, \tau)$ is the state transition matrix of $A(t)$.
- Numerical Computation (Differential Equation):
  To practically compute W_c numerically and avoid explicitly calculating the integral and state transition matrix, we can solve its differential equation form:

$$\dot{W}_c(t) = A(t)W_c(t) + W_c(t)A(t)^\top + B(t)B(t)^\top$$

- Minimum Control Energy:
  The minimum control energy E required to reach a specific state deviation x_f is given by:

$$E = x_f^\top W_c^{-1} x_f$$

E. Validation

verify terminal conditions, Hamiltonian consistency, adjoint boundary conditions, and shooting residuals associated with the TPBVP formulations.

controllable and weakly controllable linearization comparisons

- sensitivity to initial condition error
- control energy consumed
- transient behavior

IV. PART II - Optimal Trajectory Tracking

A. Phase formulation

Objective:

$$J = \int_0^{t_f} \Big[(x - x_{\text{ref}})^\top Q (x - x_{\text{ref}}) + u^\top R u\Big] dt + (x(t_f) - x_{\text{ref}}(t_f))^\top Q_f (x(t_f) - x_{\text{ref}}(t_f)) \tag{2}$$

B. Analytical Derivations (PMP)

- The Hamiltonian:

$$H = (x - x_{ref})^\top Q (x - x_{ref}) + u^\top R u + p^\top (Ax + Bu)$$

- Adjoint Equation (Costate Dynamics):

$$\dot{p} = -\frac{\partial H}{\partial x}$$

$$\dot{p} = -2Q(x - x_{ref}) - A^\top p$$

- Stationarity (Optimal Control Law):

$$\frac{\partial H}{\partial u} = 0 \implies 2Ru + B^\top p = 0$$

$$u^* = -\frac{1}{2} R^{-1} B^\top p$$

- Transversality (Terminal Condition):

$$p(t_f) = \frac{\partial \phi}{\partial x}\bigg|_{t=t_f}$$

$$p(t_f) = 2Q_f(x(t_f) - x_{ref}(t_f))$$

- Tracking Riccati Equations (Riccati substitution)
  The added auxiliary vector $s(t)$ accounts for the tracking offset

$$p(t) = P(t)x(t) + s(t)$$

$$\dot{p} = \dot{P}x + P\dot{x} + \dot{s}$$

  substitute $\dot{p}$ and $\dot{x}$

$$-2Q(x - x_{ref}) - A^\top (Px + s) = \dot{P}x + P(Ax + Bu^*) + \dot{s}$$

  substitute u ($u^* = -\frac{1}{2}R^{-1}B^\top(Px + s)$)

$$-2Qx + 2Qx_{\text{ref}} - A^\top Px - A^\top s = \dot{P}x + PAx - \tfrac{1}{2}PBR^{-1}B^\top(Px + s) + \dot{s}$$

  group the terms by $x(t)$

$$\Big[\dot{P} + PA + A^\top P - \tfrac{1}{2}PBR^{-1}B^\top P + 2Q\Big]x(t) + \Big[\dot{s} + A^\top s - \tfrac{1}{2}PBR^{-1}B^\top s - 2Qx_{\text{ref}}\Big] = 0 \tag{3}$$

  Since the expression must hold true for any arbitrary trajectory $x(t)$, we get the following tracking Riccati pair

  - Matrix Riccati Differential Equation (The Feedback Rulebook)

$$\dot{P} = -PA - A^\top P + \frac{1}{2}PBR^{-1}B^\top P - 2Q$$

  - Auxiliary Vector Differential Equation (The Feedforward Tracker)

$$\dot{s} = -\left(A^\top - \frac{1}{2}PBR^{-1}B^\top\right)s + 2Qx_{ref}$$

- Closed-Loop Tracking Control Law
  Once you integrate both $\dot{P}$ and $\dot{s}$ backward from t_f to 0, you can construct your final control law. Substitute $p(t) = P(t)x(t) + s(t)$ back into u:

$$u^*(t) = -\frac{1}{2}R^{-1}B^\top P(t)x(t) - \frac{1}{2}R^{-1}B^\top s(t)$$

  This is elegantly split into two components:

$$u^*(t) = -K(t)x(t) + u_{ff}(t)$$

  $K(t)x(t)$ is the feedback component that stabilizes the rocket and kills deviations. $u_{ff}(t)$ is the feedforward component that anticipates the c

C. Analysis

- Tracking Performance
- Transient Response
- Control Effort
- Robustness to deviations from the nominal trajectory.

What if the reference trajectory is not known in advance? Your discussion must explain what challenges does your current approach face and why.

D. Validation

V. PART III - THE MISSION PROBLEM

A. Phase A — Minimum-Time Ascent

1) Formulation:

$$J_A = \int_0^{t_1} 1\, dt = t_1$$

2) Analytical Derivations (PMP): Hamiltonian

$$H_A(x,u,p) = 1 + p^\top (Ax + Bu) = 1 + p^\top Ax + p^\top Bu$$

Adjoint equation

$$\dot{p} = -\frac{\partial H_A}{\partial x} = -A^\top p$$

Switching Function

$$\frac{\partial H_A}{\partial u} = B^\top p$$

$$S(t) = B^\top p(t)$$

$$u_i^*(t) = \begin{cases} U_{max,i} & \text{if } S_i(t) < 0 \\ U_{min,i} & \text{if } S_i(t) > 0 \end{cases}$$

$$H_A(t) \equiv 0 \implies 1 + p(t)^\top \big(Ax(t) + Bu^*(t)\big) = 0 \quad \forall t \in [0, t_1]$$

3) Implementation & Ascent Trajectory Results:

4) Validation:

B. Phase B — Return-and-Land Problem

Phase B begins precisely where Phase A ends, thus the initial condition for the landing phase is $x(0) = x_1$

1) Formulation & Core PMP:

$$J_B = \int_0^{t_f} \Big(x^\top Qx + u^\top Ru\Big) dt$$

Generic Hamiltonian, and the fundamental $H(t_f) = 0$ condition)

$$H_B(x,u,p) = x^\top Qx + u^\top Ru + p^\top (Ax + Bu)$$

The unconstrained control law and adjoint equations remain identical to Phase I:

$$u^*(t) = -\frac{1}{2} R^{-1} B^\top p(t)$$

$$\dot{p}(t) = -2Qx(t) - A^\top p(t)$$

Because t_f is free, PMP imposes the free-time transversality condition. The Hamiltonian evaluated at the terminal time must equal zero:

$$\begin{aligned} H_B(t_f) = 0 \implies & x(t_f)^\top Qx(t_f) + u(t_f)^\top Ru(t_f) \\ & + p(t_f)^\top \big(Ax(t_f) + Bu(t_f)\big) = 0 \end{aligned}$$

2) Transversality:

$$p(t_f) = \frac{\partial \phi}{\partial x}\bigg|_{t_f} + \sum_j \nu_j \nabla h_j(x(t_f))$$

- Manifold 1 (Pinpoint Landing)
  - Description
$$h_1(x) = p_z = 0$$
$$h_2(x) = v_z = 0$$
$$h_3(x) = \theta = 0$$
  - Gradients
$$\nabla h_1 = [0,1,0,0,0,0,0]^\top$$
$$\nabla h_2 = [0,0,0,1,0,0,0]^\top$$
$$\nabla h_3 = [0,0,0,0,1,0,0]^\top$$
  - Transversality
$$p_{p_x}(t_f) = \frac{\partial \phi}{\partial p_x} \quad \text{(Free state)}$$
$$p_{p_z}(t_f) = \frac{\partial \phi}{\partial p_z} + \nu_1 \quad \text{(Constrained)}$$
$$p_{v_x}(t_f) = \frac{\partial \phi}{\partial v_x} \quad \text{(Free state)}$$
$$p_{v_z}(t_f) = \frac{\partial \phi}{\partial v_z} + \nu_2 \quad \text{(Constrained)}$$
$$p_\theta(t_f) = \frac{\partial \phi}{\partial \theta} + \nu_3 \quad \text{(Constrained)}$$
$$p_\omega(t_f) = \frac{\partial \phi}{\partial \omega} \quad \text{(Free state)}$$
$$p_m(t_f) = \frac{\partial \phi}{\partial m} \quad \text{(Free state)}$$
- Manifold 2 (Surface Landing)
  - Description: landing on the surface of a dome/sphere.
$$h_1(x) = (p_x - p_c)^2 + p_z^2 - r^2 = 0$$
$$h_2(x) = v_z = 0$$
$$h_3(x) = \theta = 0$$
  - The Gradients
    h_1 couples p_x and p_z, spanning 2 dimensions:
$$\nabla h_1 = [2(p_x - p_c), 2p_z, 0,0,0,0,0]^\top$$
$$\nabla h_2 = [0,0,0,1,0,0,0]^\top$$
$$\nabla h_3 = [0,0,0,0,1,0,0]^\top$$
  - Transversality
$$p_{p_x}(t_f) = \frac{\partial \phi}{\partial p_x} + 2\nu_1 (p_x(t_f) - p_c)$$
$$p_{p_z}(t_f) = \frac{\partial \phi}{\partial p_z} + 2\nu_1 p_z(t_f)$$
    (The remaining 5 costates are identical to M1)
  - Conclusion for M2: We can eliminate the unknown multiplier ν_1 by dividing the first two equations (assuming $\frac{\partial \phi}{\partial x} = 0$). This yields the exact boundary condition passed to the TPBVP solver:
$$\frac{p_{p_x}(t_f)}{p_{p_z}(t_f)} = \frac{p_x(t_f) - p_c}{p_z(t_f)}$$

3) Landing Manifold Comparison:

4) Validation:

VI. Part IV - The Nonlinear Optimal Control Near Hover

A. Phase Formulation

Consider the fixed-time, free-endpoint nonlinear optimal control problem

$$\min_u \int_0^{t_f} \left(x^\top Q x + u^\top R u\right) dt$$

Recall:

$$\dot{x} = \begin{bmatrix} v_x \\ v_z \\ -\frac{T}{m}\sin\theta \\ \frac{T}{m}\cos\theta - g \\ \omega \\ \frac{\tau}{I} \\ -\alpha T \end{bmatrix}$$

B. Analytical Derivations

- Nonlinear Hamiltonian Adjoin the nonlinear dynamics to the running cost using the costate vector $p = [p_1, p_2, p_3, p_4, p_5, p_6, p_7]^\top$:

$$H(x,u,p) = x^\top Q x + u^\top R u + p_1 v_x + p_2 v_z - p_3 \frac{T}{m}\sin\theta + p_4 \Big(\frac{T}{m}\cos\theta$$

- Adjoint equations ($\dot{p} = -\frac{\partial H}{\partial x}$)

$$\begin{aligned}
\dot{p}_1 &= -2Q_{11}p_x \\
\dot{p}_2 &= -2Q_{22}p_z \\
\dot{p}_3 &= -2Q_{33}v_x - p_1 \\
\dot{p}_4 &= -2Q_{44}v_z - p_2 \\
\dot{p}_5 &= -2Q_{55}\theta + p_3\frac{T}{m}\cos\theta + p_4\frac{T}{m}\sin\theta \\
\dot{p}_6 &= -2Q_{66}\omega - p_5 \\
\dot{p}_7 &= -2Q_{77}m - p_3\frac{T}{m^2}\sin\theta + p_4\frac{T}{m^2}\cos\theta
\end{aligned}$$

- Stationarity conditions ($\frac{\partial H}{\partial u} = 0$)
  For Thrust (T):

$$\frac{\partial H}{\partial T} = 2R_{11}T - p_3\frac{\sin\theta}{m} + p_4\frac{\cos\theta}{m} - p_7\alpha = 0$$

$$T^* = \text{sat}_{T_{min}}^{T_{max}}\left[\frac{1}{2R_{11}}\left(p_3\frac{\sin\theta}{m} - p_4\frac{\cos\theta}{m} + p_7\alpha\right)\right]$$

  For Torque (τ):

$$\frac{\partial H}{\partial \tau} = 2R_{22}\tau + \frac{p_6}{I} = 0$$

$$\tau^* = \text{sat}_{\tau_{min}}^{\tau_{max}}\left[-\frac{p_6}{2IR_{22}}\right]$$

- TPBVP
- state equations, costate equations
- terminal conditions, and optimality conditions

C. Numerical Implementation

D. Analysis

1. the linear optimal controller on the linearized dynamics
2. the same controller on the nonlinear dynamics
3. and the nonlinear optimal controller on the nonlinear dynamics.

- trajectory comparisons (position, velocity, attitude, and mass trajectories) control histories (thrust and torque histories)
- Hamiltonian evolution
- Convergence behavior
- Solver sensitivity
- Terminal residual analysis (terminal error comparisons)

E. Validation

Verify terminal conditions Hamiltonian consistency Adjoint boundary conditions Shooting residuals associated with the TPBVP formulations.

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REFERENCES

[1] S. Coyle, E. Rouse, and C. Majidi, "Actuation and design innovations in earthworm-inspired soft robots: A review," *Frontiers in Robotics and AI*, vol. 10, 2023. [Online]. Available: `https://pmc.ncbi.nlm.nih.gov/articles/PMC9989016/`